

Economic Benefits of Cross Border Power Trading: A Case Study for Bangladesh

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Abstract— Bangladesh is already importing power from its neighboring country India via Cross Border Electricity Trade (CBET) arrangements. Such trading could be expanded rapidly to reap major benefits *inter alia* a reduction in expensive oil-based generation, less rental peaking capacity, unserved energy and CO₂ emissions. Trade will also help India to manage its own peak and thereby lower volatility in spot prices. This study analyzes benefits of an additional 1 GW power trading via Bangladesh-India interconnector through a market-driven mechanism to assess these benefits. A short-term dispatch model is used that uses past spot prices in the Indian Energy Exchange (IEX) to drive import/export decisions and calculate the benefits to both countries. Annual benefits are estimated to be in the range of USD 90-286 million that suggests a very low payback period of such investments.

Keywords— Cross Border Power Trade, Indian Energy Exchange, Dispatch optimization.

I. INTRODUCTION

Pressure on electricity costs, energy security and rising emissions from fossil-fuel use are among the key concerns in the energy sector of the South Asian countries i.e. Bangladesh, Bhutan, India, Nepal and Sri Lanka. These countries fall below their aspirations for the higher levels of per capita electricity consumption that can sustain recently rapid economic expansion. This is a significant concern for a region that is home to more than 1.8 billion people.

Diversified CBET between Bangladesh, Bhutan, India, Nepal, and perhaps in future Sri Lanka, holds a key to meet accelerating electricity demand in this region with lower cost generation and emissions [1]. The benefits of CBET are being recognized gradually between Bangladesh and India, Nepal and India, and Bhutan and India. Affordable, reliable and cleaner electricity supplied through CBET for Bangladesh will require unlocking access to hydropower from neighboring countries like Nepal, Bhutan and North-Eastern part of India. A positive sign towards unleashing this CBET potential is that countries in this region are discussing new interconnections and how to optimize cross border power trading to extract mutual benefits.

Successful CBET among countries can promote greener and more cost-effective grids that enable consideration of alternatives the overreliance on fossil-fuel based generation options. Chattopadhyay and Fernando [2] had estimated in 2011 that CBET in South Asia can reduce 28 million tonnes of CO₂ emissions annually by displacing thermal generation in India with hydro from Nepal/Bhutan. The advent of solar and wind in the region since then has also significantly enhanced such reduction potential. The integration of large-scale renewable resources is also being taken into account and received

considerable attention in recent years in other parts of the world [2]. However, not all countries or regions have sufficient resources or land to make these transformations. While each project must be evaluated on its own merits, various CBET studies suggest overall benefits of CBET in South Asia are significant. Some of the recent studies such as the SAARC Regional Energy Trade Study (SRETS)[3], the Vision 2020[4] and Programme on “Energy Trade between SAARC Countries” (PENT) [5] and studies by the World Bank [6] and Asian Development Bank(ADB)[7] have identified many opportunities and requirements for power trading in the South Asia region. All of these studies have established an overwhelming economic case for CBET in South Asia. The World Bank study [8], for instance, estimated total benefit of (unconstrained) trading over the next 20 years at \$100 billion. The ADB study also estimated near-term potential for electricity trading among SAARC Member States to save at least \$1 billion annually and promoted the concept of a Regional Power Exchange (RPX).

The present research study is a complement to previous studies [1,7,8] with a market-based analysis of CBET that has not been conducted in the past. It provides a fresh perspective into the near-term benefits of trade under a range of realistic scenarios that would shape volume, direction and benefits of trade. The scope of this study is limited to the interconnection options between Bangladesh and India. However, it may act as a guide for planners in Bangladesh to encompass such interconnection prospects along with other energy efficiency and demand response measures [9]. There may need to be an operational process by which the Single Buyer (on a limited context) or the generation contracting agency (e.g., Bangladesh Power Development Board) can actively implement such decisions closer to time [9]. This is where the short-term operational analysis taking into consideration the uncertainties around demand growth, has a role to play.

A. Bangladesh Context

Bangladesh has made significant strides to more than triple its power generation capacity from 5+ GW to 18+ GW in the last decade. (Figure 1). However, as much as 6 GW or about one-third of the presently installed capacity is fueled by liquid fuel based (HFO or diesel) capacity. These power plants were developed as a stop-gap measure to meet the peak demand. Peak demand has already been close to 13 GW and is projected to have sustained growth leading towards 21+ GW by 2025, 29+ GW by 2030 and 55+ GW by 2041, as per the Power System Master Plan[10]. Reliance on oil has soared more than 22% of the generation in 2017/18 which costs the country about USD 2 billion annually [11]. Further, as domestic gas reserve

declines, increasing generation demand may increase the requirement of liquid fuels. Although Bangladesh is gradually phasing out such dependency on liquid fuel based generation.

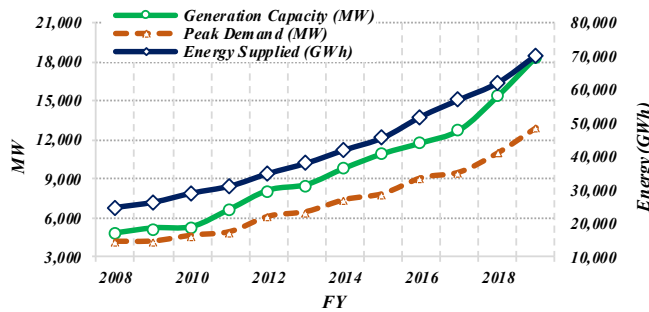


Figure 1: Annual peak demand, generation capacity, and energy supplied; Bangladesh

Bangladesh is currently adopting optimization-based planning measures to decide optimum primary fuel mix since domestic gas is reported to be noticeably reduced in the foreseeable future.

Bangladesh established its first electricity interconnection (500 MW) with the neighboring country India in October 2013. The project was financed by ADB. This interconnection was proven efficacious to feed a portion of rapidly growing power demand in Bangladesh and also the political commitment of the two governments to proceed with cross border power trading. A back-to-back HVDC transmission system arrangement has overcome the technical difficulties in the system control with 220 GW+ Indian power system being connected with comparatively small 11 GW system of Bangladesh at the time. The interconnection allowed Bangladesh to access the power market in India for a capacity of 250 MW through a commercial agreement while the remaining 250 MW is supplied through a set of Government of India owned power plants[7]. Later Bangladesh has extended its import scope via other contracts and at present, Bangladesh is importing 1.16 GW of power from India with the projection of extending the import up-to additional 1.9+ GW by 2023. GoB is in discussion regarding CBET with Bhutan and Nepal vis-à-vis the 2 GW interconnection in near future[12].

Larger scale power transfers and tighter integration among power systems would eventually generate needs for greater efficiency in market and system operations. More interest from various suppliers and buyers in CBET would lead to a need for increasing reliance on open competition to obtain efficient pricing. As the regional power market evolves from bilateral to multi-country and multi-company transactions, in larger volumes and varied terms, some regulatory alignment may be necessary and could lead to a discussion on developing a regional power pool. Power traded through market-driven prices may further build investors' confidence in trade-related generation and transmission assets. While political, commercial and technical risks will remain, continued efforts to align market and system operations efficiently and transparently can help to further strengthen allocation and mitigation of residual risks in CBET.

B. Key Issues

The objective of this dispatch analysis is to assess the short-term CBET prospects between Bangladesh and India. The analysis starts in the context that the rapidly growing demand in Bangladesh may trigger thought-provoking demand-supply scenarios. The analysis estimates a range of short-term benefits generated from an additional 1 GW of interconnection capacity between Bangladesh and India which is reported to be under planning. This what-if analysis includes the assumed fully operational interconnection and that CBET occurs based on IEX spot prices. CBET is bi-directional, i.e. DISCOMs and generators in Bangladesh may need to have access to IEX. The analysis leaves aside commercial and regulatory issues, that can be addressed through bulk power transmission agreements and power purchase agreements. Several other factors have an influence on the results of this analysis. Among the critical ones are gas constraints in Bangladesh, gas pricing, and the IEX spot prices. Such factors need to be understood to have an influence on the variability of the benefits presented in this study. These factors were accounted for in five "Scenarios" for the analysis shown below.

II. MODELLING METHODOLOGY

We have undertaken a detailed modeling study based on EPM[13] to assess the interconnector benefits within the scope of economic analysis. This analysis contains the following analytical components underlying the model:

- Hourly dispatch model is used to meet the demand for each hour of 2017/18 in Bangladesh (divided into two zones – East and West with a transfer limit of 1,200 MW at the high end). A domestic gas supply limit is imposed on an annual basis that at the low end of 1000 mmcf/d can severely restrict the ability to use the cheaper domestic gas-based generation facilities. Although the historic gas allocation to the power sector has been at 1000 mmcf/d, there has been higher allocation up to 1200 mmcf/d and further potential to go up to 1400 mmcf/d that would significantly increase gas generation potential. In our analysis, we consider all three gas limits in the range of 1000-1400 mmcf/d and examine its ramification for cross-border trade.
- Bangladesh has the option in the trade scenario to buy and sell from the IEX at Area Clearing Price (ACP) of up to 1,000 MW.
- A key assumption is that in this analysis is- Bangladesh sells and buys in the "residual" market of the Day Ahead Market (DAM) in India. That is, trading will take place at the margin of ACP after IEX has cleared -- i.e., demand in the Indian market is met first in the model. Import from India, or export to India, happens at the given market clearing prices. Bangladesh puts supply offers and demand bids in IEX. ACP for E1 (Eastern Region)[14]- has been considered for the analysis. Price elasticity is approximated from the slope of the supply curves in DAM.

The model provides valuable insights into what may be the near-term benefits of market driven trading CBET using real prices. If these benefits transpire to be significant, it is an obvious indication of the attractiveness of the interconnector(s).

Economic benefits of CBET are expressed in one or more of three major forms, namely:

- Reliability benefit:** Avoiding the cost of unserved energy to the extent that the new transmission asset lowers peak and energy curtailment in the importing region;
- Operating cost benefits:** Avoiding the cost of expensive primary fuel (HSD/HFO) and O&M costs that the power flows on the new interconnection line may displace; and
- Capacity benefits:** Any avoided cost of new generation capacity that the interconnection line renders unnecessary, albeit the dispatch analysis presented here does not capture this.

III. CASE STUDY FOR BANGLADESH

The EPM short-term model and the dispatch model consider more than 130 generating units (existing) totaling around 17.6 GW generation capacity. The peak demand is close to 13 GW. The merit order curve for existing generation capacity with cost assumptions based on historical opex is shown in Figure 2. It should be noted that the generation potential from low cost domestic gas capacity (below \$50/MWh) is limited by gas availability.

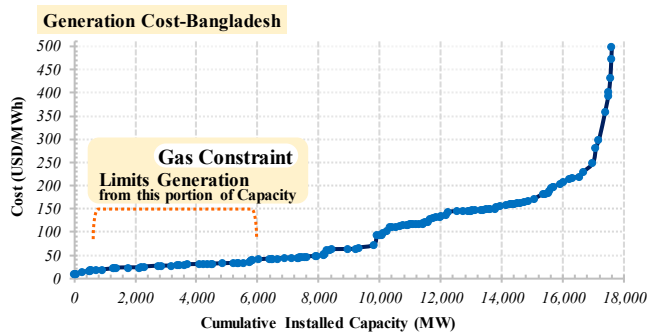


Figure 2: Generation cost for the Installed Capacity in Bangladesh

Figure 3 shows the historic prices on the IEX. There are some important insights on trade potential that can be derived from Figures 2 and 3. For any demand level in Bangladesh crossing over 6,000 MW, the marginal cost rises above USD 40/MWh for Bangladesh. It clearly indicates that even without accounting for gas availability limits – there is room for import for Bangladesh in several thousand hours.

Alternatively, in the year 2017/18 IEX prices peak in India more than USD 80/MWh (Figure 3) for more than 530 hours. It indicates the benefits for Bangladesh to export to India to lower overall system costs.

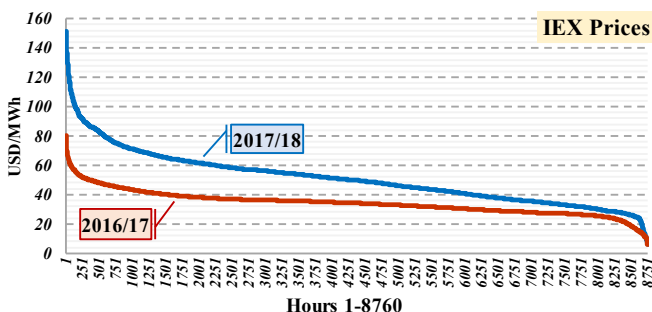


Figure 3: IEX ACP for the last two years

Existing interconnection with India facilitates 1.16 GW of imports for Bangladesh (shown in TABLE I). The addition of another 1.9 GW import for Bangladesh is expected to be completed by the first quarter of 2022 (shown in TABLE II).

TABLE I: EXISTING INTERCONNECTORS WITH INDIA

Sl.	Link	Description	Cost (mill. USD)	Fund
1.	Tripura	Phase-I (100 MW) - Mar, 2016	21	GoB & PGCB
		Phase-II (60 MW) - Sep, 2017		
2.	Bheramara HVDC	Phase-I (500 MW) - Oct, 2013	197	ADB, GoB & PGCB
		Phase-II (500 MW) - Sep, 2018	232	

Note: Phase-I & II are obliged under different contracts, terms & conditions and tariff rates. Total project cost collected from Audited Financial Reports by ADB[15]. GoB, PGCB denotes the Government of Bangladesh and Power Grid Company of Bangladesh respectively.

TABLE II: ONGOING CROSS BORDER PROJECTS WITH INDIA

Sl.	Link	Description	Fund
3.	Tripura	Phase-III (340 MW) - Mar, 2021	ADB (expected*), GoB & PGCB
4.	Adani Power	1,600 MW - Dec, 2022	GoB & PGCB

**Fund is not yet confirmed from ADB. Note: Tentative completion year has been mentioned.*

1) Short-term analysis

Short term analysis encompasses the total benefits from the gain of trade for the two systems altogether. It includes savings prospects that Bangladesh and India both may make through imports. The results extracted from this analysis represent purely opex savings (based on primary fuel costs). Economic benefits for each country are attributed from the difference in (generation with and without cross border trade) on an hourly basis multiplied by (generation cost less IEX price). Therefore, benefits denote the savings made by each country from its respective import volume that replaces more expensive generation.

2) Scenarios to Assess the Benefits of Interconnector

The following assumptions have been made to develop the scenarios for analysis-

- Domestic gas constraints for Bangladesh has been varied from 1000 mmcf/d at the low end to 1200 mmcf/d (average) and 1400 mmcf/d at the high end. IEX clearing Prices for 2017/18 has been considered as high price and 2016/17 as average/low prices.
- Current Gas Prices (CGP) used for dispatch and a higher rate- 'market'/economic gas price (EGP) has also been considered as domestic gas price in Bangladesh.
- Base Case: IEX price for Year 2017/18 with the current gas price and domestic gas supply of 1200 mmcf/d. We assumed that it represents the gas price/supply condition in the most recent year. S1-S4 (as shown in Table III): developed on the basis of gas limits, IEX price year and CGP vs EGP.

3) Short term Analysis Results

a) Economic Analysis

Overall **gross** benefits have been found significant, which ranges of USD 90-286 million for a single year. The existing Bangladesh-India interconnection project (of 500MW) that commissioned in 2013 had a total cost of USD 197 million. It is therefore likely that the incumbent 1 GW project would have a very favorable payback period and return on investment.

Gross benefits even in the worst case (based on gas availability and high IEX price) is significant.

TABLE III: KEY RESULTS FROM ECONOMIC ANALYSIS

Sl.	Scenario	Total Benefits (million USD)	BD (GWh)		Benefits (million USD)	
			Import	Export	BD	IND
1.	Base 1200 mmcf/d at CGP	137	6,141	1,316	75	61
2.	S1 1400 mmcf/d at CGP	90	4,023	2,865	23	67
3.	S2 1000 mmcf/d at CGP	286	7,363	581	222	63
4.	S3 1200 mmcf/d at EGP	116	5,897	1,486	97	19
5.	S4 1000 mmcf/d at EGP	218	8,656	9	218	0.1

Benefit sharing for a single year (as shown in Figure 4) for each country based on the opex savings may vary widely depending on the key variables, including but not limited to the IEX price, gas constraints, and interconnector capacity.

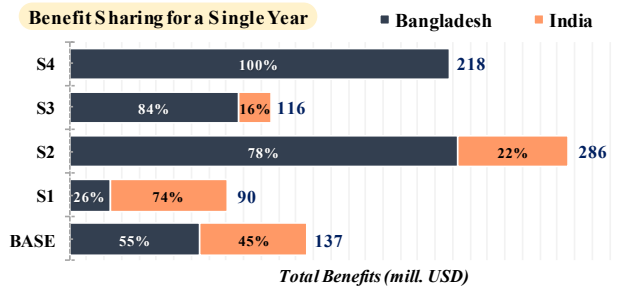


Figure 4: Benefits sharing found in Economic Analysis based on the realized value for a single year

If the cheaper domestic gas in Bangladesh is in short supply (up-to 1000 mmcf/d); the overall benefits of the interconnector link would substantially increase more than three times. A major portion of the benefits could be attributed to Bangladesh, as presented in scenario S2. Maximum benefits could be generated in the scenario when in addition to low gas availability in Bangladesh, IEX prices is also lower (e.g., 2016/17 level as shown in Figure 3). The EGP scenarios (S3 & S4) suggest that even at market fuel prices, benefits would remain significant and largely attributed to Bangladesh, even under high IEX price cases.

b) Monthly Trading for Bangladesh

Significant variability in import and export has been found. Imports for Bangladesh would have been beneficial during Jul-Aug'17 and also for Jun'18 (benefit sharing shown in Figure 7). Exports have been found prominent in low demand months in Bangladesh.

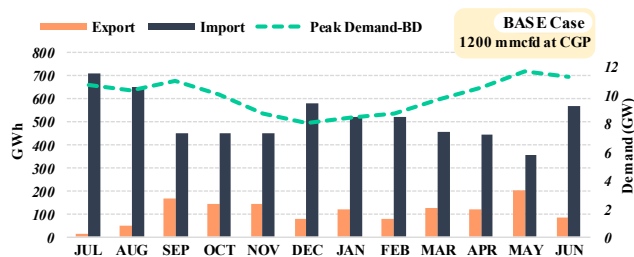


Figure 5: Monthly Trading for Bangladesh – Base Case

Gas availability alters the pattern of power flows significantly by changing the direction firmly towards Bangladesh even if IEX prices are high.

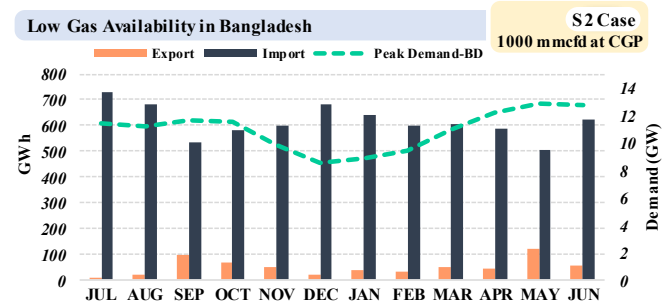


Figure 6: Monthly Trading (GWh) for Bangladesh– S2 Case (low gas availability in Bangladesh)

c) Benefits for both Bangladesh & India

Significant benefits have been produced for every individual month and indicate a well-balanced, mutual, and significant benefit yield from bidirectional short-term trade, barring the low gas availability scenario. Base Case results show that (Figure 7) imports by Bangladesh would have been very beneficial during Jul-Aug'17 and Jun'18 which are the peak demand months in the country. Exports to India, however, are far more beneficial for a wide range of Sep'17 to May'18 due to the high IEX prices. Therefore, Bangladesh could extract a higher share of the benefits by exporting to India while the in-country demand is comparatively lower than that of the peak season.

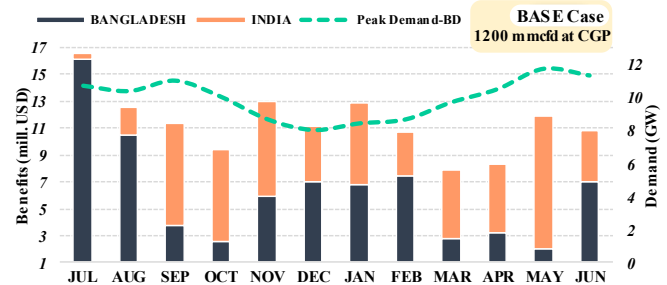


Figure 7: Monthly Benefits for both Bangladesh & India – BASE Case

Massive benefits for all months accrue to Bangladesh under low gas availability (Figure 8). On the contrary, exports from India's perspective also yield quite significant benefits especially around Sep'17-Oct'17 and Mar-May'18; although, are much lower than that of Bangladesh.

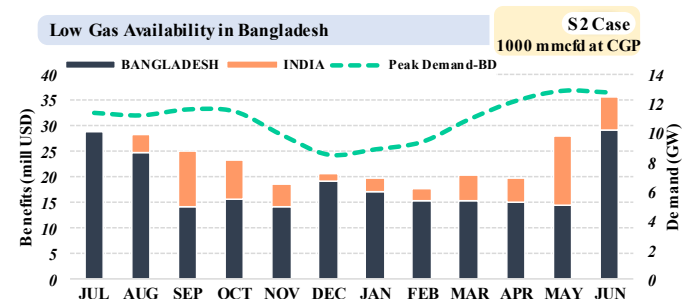


Figure 8: Monthly Benefits for both Bangladesh & India – S2 Case (low gas availability in Bangladesh)

d) Import-Export Flow Analysis

Figure 9 shows the cumulative flow duration curve, i.e., the number of hours (x-axis) at a specific flow level (y-axis) which is capped at 1000 MW indicating the line is congested for a certain number of hours. Bangladesh imports about 6,141 GWh with the link congested for more than 4,000 hours whereas, Bangladesh exports 1,316 GWh during peak prices in India with very high value and the link is also constrained in that direction for more than 600 hours.

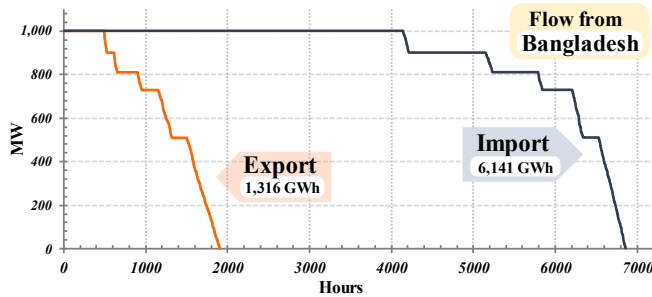


Figure 9: Import/Export flow for both Bangladesh & India – Base Case

e) Cost Recovery for Transmission Interconnector

Investment recovery mechanisms in transmission interconnection are in practice based on costs of transmission investment. The approach to passing these costs through to consumers is a regulatory decision based on a range of considerations, including understanding who benefits from the investment. Additional transmission lines can be justified based on a number of considerations and as the regional market grows these may include multi-country CBET. This study focuses on benefits accruing to a case of bidirectional, bilateral trade on 1 GW of additional transfer capacity. Benefits are likely to be different, and larger, if the bilateral constraint in the model are redefined to take into account a larger number of CBET countries and transactions. There are multiple business models that can support a scaled up, win-win CBET. Additional analysis following this study presented here are planned. Additional contributions would be useful to inform further discussions on the subject.

IV. CONCLUDING REMARKS

- A short-term analysis using a price-driven dispatch model that captures only opex savings from cross-border trade provides comfort that the benefits of a 1 GW link between Bangladesh and India found significant enough to strongly justify the investment in that interconnection facility.
- Even with the relatively high IEX prices, such as those that that occurred in year 2017/18, the model results show that the major share of the value of the link could be extracted (more than USD 100 m. in the Base Case) in a single year, which suggests a very favorable payback period for such a project.
- The modelling results indicate that the 1 GW interconnector would be beneficial to both countries and that the magnitude and distribution of benefits depend on a number of factors including domestic gas price and availability, and the IEX price variability. While the magnitude and distribution of benefits also vary seasonally, as demonstrated by the month-

by-month modelling results, the aggregate benefits are noticeably significant.

- It is recognized that some market and operating rules and settlement mechanisms, particularly for participation in the Indian power exchanges, need to be put in place, piloted and tested. In addition, each of the trading countries, including but not limited to Bangladesh, will need to work to align regulatory environments to support growth in mutually beneficial CBET.

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